LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Classification of aircraft, arrangements of major components and their functions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Classification of aircraft, arrangements of major components and their functions
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of component arrangement diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary functions of an aircraft structure?
(b) Can you list three major components of an aircraft?
(c) How is an aircraft generally classified based on its configuration?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of aircraft by configuration • Classification by propulsion type • Classification by mission role • Classification by size and capacity • Classification by wing configuration	L & Ds	PPT, IB	15 min
(b)	• Arrangement of fuselage and cockpit • Wing structure and control surfaces • Empennage components and functions • Landing gear types and mechanisms • Powerplant location and integration	L & Ds	PPT, PDF	20 min
(c)	• Structural elements: skin, frames, and ribs • Load distribution and stress analysis • Material selection for airframe • Fastening methods and joints • Corrosion protection and maintenance access	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between monoplane and biplane configurations.
(b) Describe the function of the horizontal stabilizer in the empennage.
(c) How does the landing gear arrangement affect aircraft performance?

12. Home Assignment and Follow Up	Prepare a brief note on the design requirements for the next lesson (LP 02) focusing on how classification influences manufacturing processes.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Steps in aircraft design, manufacturing process and aircraft design requirements	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min